Lemma：E和F是域，$f: E \to F$是环同态，则有：$f(E)$是F的子域.

Proof：$\because$ E和F是环，$f: E \to F$是环同态 $\therefore f(E)$是F的子环.

$\therefore 0_F, 1_F \in f(E)$.

对$\forall a, b \in f(E)$. $\because f(E)$是F的子环 $\therefore -b \in f(E)$

$\therefore a+(-b) \in f(E)$ $\therefore a-b = a+(-b) \in f(E)$

对$\forall a \in f(E)$，$\forall b \in f(E) \setminus \{0_F\}$，有：

$\because b \in f(E)$ $\therefore \exists \lambda \in E$，s.t. $b = f(\lambda)$

假设$\lambda = 0_E$，则$b = f(\lambda) = f(0_E) = 0_F$

$\because b \in f(E) \setminus \{0_F\}$ $\therefore b \notin \{0_F\}$ $\therefore b \neq 0_F$ 矛盾.

$\therefore \lambda \neq 0_E$ $\therefore \lambda \in E \setminus \{0_E\} = E^\times$ $\therefore \lambda^{-1} \in E^\times$，$\lambda^{-1} \in E$.

$\because f(\lambda^{-1}) = (f(\lambda))^{-1}$ $\therefore f(\lambda^{-1}) = b^{-1}$

$\therefore b^{-1} = f(\lambda^{-1}) \in f(E)$ $\therefore ab^{-1} \in f(E)$

$\therefore f(E)$是F的子域. $\square$

Recall lemma：对$\forall N\in\mathbb{Z}_{\geq 1}$，有：$\mathbb{Z}/N\mathbb{Z}$是域 $\Longleftrightarrow$ N是素数

定义（域$\mathbb{F}_p$） 对$\forall$素数p，$\mathbb{Z}/p\mathbb{Z}$是域，记作 $\mathbb{F}_p:=\mathbb{Z}/p\mathbb{Z}$

Remark：$\mathbb{F}_p=\mathbb{Z}/p\mathbb{Z}=\{[x]_p \mid x\in\mathbb{Z}\}=\{[0]_p,[1]_p,[2]_p,\cdots,[p-1]_p\}$

以上内容来自3.3多项式环(5)笔记.

Recall lemma：R是一个任意的环，则有：存在唯一的环同态$\mathbb{Z}\to R$.

$f:\mathbb{Z}\longrightarrow R$ 就是 $\mathbb{Z}\to R$的唯一的环同态.
$\quad x\longmapsto x\cdot 1_R$

以上内容来自3.7域的特征(1)笔记.

Recall lemma：R是整环，则有：$\mathrm{char}(R)=0$或$\mathrm{char}(R)$是素数.

3.7域的特征(1)笔记已证.

Recall lemma：F是域，R是非零环，映射$\varphi:F\to R$是环同态，则有：映射φ是单射.

3.2同态和同构(4)笔记已证.

Recall lemma：$(\mathbb{Z},+,\cdot,0,1)$是一个整环，$\mathrm{Frac}(\mathbb{Z})=\mathbb{Q}$是域.

3.5(3)，2.6(5)笔记已证.

lemma：F是一个任意的域，则有：存在唯一的环同态$\mathbb{Z}\to F$.

Proof：$\because F$是域 $\therefore F$是环 $\therefore$存在唯一的环同态$\mathbb{Z}\to F$.

$\eta_F:\mathbb{Z}\longrightarrow F$ 就是 $\mathbb{Z}\to F$的唯一的环同态 $\square$
$\quad x\longmapsto x\cdot 1_F$

Lemma：F是一个域且 $\mathrm{char}(F)=0$，则有：$\eta_F: \mathbb{Z} \to F$ 是单同态.
$x \mapsto x\cdot 1_F$

Proof：$\because F$是域 $\therefore$已经证明了 $\eta_F: \mathbb{Z} \to F$ 是 $\mathbb{Z}\to F$的唯一的环同态
$x \mapsto x\cdot 1_F$

对$\forall x, y \in \mathbb{Z}$，若$\eta_F(x)=\eta_F(y)$，则有：$x\cdot 1_F = y\cdot 1_F$

$\because x, y \in \mathbb{Z}$ $\therefore x-y \in \mathbb{Z}$

$\therefore (x-y)\cdot 1_F = (x+(-y))\cdot 1_F = x\cdot 1_F + (-y)\cdot 1_F \overset{\text{对}\forall y\in\mathbb{Z}\text{成立}}{=} y\cdot 1_F + (-(y\cdot 1_F))$

$= 0_F$ $\therefore \mathrm{char}(F) \mid (x-y)$. $\because \mathrm{char}(F)=0$ $\therefore x-y=0$

$\therefore x=y$ $\therefore \eta_F: \mathbb{Z} \to F$ 是单同态. $\square$
$x \mapsto x\cdot 1_F$

现在来应用 3.5(5) 的这性质表述.

$\mathbb{Z}$是整环，$\mathbb{Q}=\mathrm{Frac}(\mathbb{Z})$是域，$f: \mathbb{Z} \to \mathbb{Q}$ 是单同态.
$x \mapsto \frac{x}{1}$

F是域且 $\mathrm{char}(F)=0$，则F是交换环，$\eta_F: \mathbb{Z} \to F$ 是单同态.
$x \mapsto x\cdot 1_F$

$\therefore \eta_F(\mathbb{Z}\setminus\{0\}) \subseteq F^\times$. $\therefore$存在唯一的环同态 $\theta: \mathbb{Q} \to F$, s.t.

下图交换：

$$\begin{tikzcd} & \mathbb{Z} \arrow[dl, "f"'] \arrow[dr, "\eta_F"] & \\ \mathbb{Q} \arrow[rr, "\theta"'] & & F \end{tikzcd}$$

定义映射：$\alpha: \mathbb{Q} \longrightarrow F$

$$\frac{a}{b} \longmapsto (b\cdot 1_F)^{-1}(a\cdot 1_F)$$

对$\forall \frac{a}{b} \in \mathbb{Q}$（其中$a\in\mathbb{Z}$，$b\in\mathbb{Z}\setminus\{0\}$），

假设 $b\cdot 1_F = 0_F$，则有 $\mathrm{char}(F) \mid b$ $\quad\because \mathrm{char}(F)=0 \quad \therefore b=0$

矛盾. $\quad\therefore b\cdot 1_F \neq 0_F \quad \therefore b\cdot 1_F \in F^{\times} \quad \therefore (b\cdot 1_F)^{-1} \in F^{\times} \subseteq F$

$\therefore \alpha\left(\frac{a}{b}\right) = (b\cdot 1_F)^{-1}(a\cdot 1_F) \in F \qquad \therefore \alpha(\mathbb{Q}) \subseteq F$

对$\forall \frac{a_1}{b_1}, \frac{a_2}{b_2} \in \mathbb{Q}$.（其中$a_1, a_2 \in \mathbb{Z}$，$b_1, b_2 \in \mathbb{Z}\setminus\{0\}$）

若$\frac{a_1}{b_1} = \frac{a_2}{b_2}$，则有：$a_1 b_2 = a_2 b_1 \in \mathbb{Z}$

$\because \eta_F: \mathbb{Z} \longrightarrow F$ 是单同态 $\quad \therefore \eta_F(a_1 b_2) = \eta_F(a_2 b_1)$

$\quad x \longmapsto x\cdot 1_F$

$\therefore \eta_F(a_1)\eta_F(b_2) = \eta_F(a_2)\eta_F(b_1) \qquad \therefore (a_1\cdot 1_F)(b_2\cdot 1_F) = (a_2\cdot 1_F)(b_1\cdot 1_F)$

$\because b_1\cdot 1_F \neq 0_F$，$b_2\cdot 1_F \neq 0_F \qquad \therefore b_1\cdot 1_F \in F^{\times}$，$b_2\cdot 1_F \in F^{\times}$

$\therefore (b_1\cdot 1_F)^{-1}(a_1\cdot 1_F) = (b_2\cdot 1_F)^{-1}(a_2\cdot 1_F)$

$\therefore \alpha\left(\frac{a_1}{b_1}\right) = \alpha\left(\frac{a_2}{b_2}\right) \qquad \therefore \alpha: \mathbb{Q} \longrightarrow F$是一个映射.

$\because \alpha\left(\frac{a_1}{b_1} + \frac{a_2}{b_2}\right) = \alpha\left(\frac{a_1 b_2 + a_2 b_1}{b_1 b_2}\right) = (b_1 b_2\cdot 1_F)^{-1}((a_1 b_2 + a_2 b_1)\cdot 1_F)$

$\because b_1, b_2 \in \mathbb{Z}\setminus\{0\} \qquad \therefore \eta_F(b_1 b_2) = \eta_F(b_1)\eta_F(b_2)$

$\therefore (b_1 b_2)\cdot 1_F = (b_1\cdot 1_F)(b_2\cdot 1_F)$

$\because a_1, b_2 \in \mathbb{Z} \quad \therefore \eta_F(a_1 b_2) = \eta_F(a_1)\eta_F(b_2)$

$\therefore (a_1 b_2)\cdot 1_F = (a_1 \cdot 1_F)(b_2 \cdot 1_F)$

$\because a_2, b_1 \in \mathbb{Z} \quad \therefore \eta_F(a_2 b_1) = \eta_F(a_2)\eta_F(b_1)$

$\therefore (a_2 b_1)\cdot 1_F = (a_2 \cdot 1_F)(b_1 \cdot 1_F)$

$$\therefore \alpha\left(\frac{a_1}{b_1} + \frac{a_2}{b_2}\right) = (b_1 b_2 \cdot 1_F)^{-1}\left((a_1 b_2 + a_2 b_1)\cdot 1_F\right)$$

$$= \left((b_1 \cdot 1_F)(b_2 \cdot 1_F)\right)^{-1}\left(a_1 b_2 \cdot 1_F + a_2 b_1 \cdot 1_F\right)$$

$$= (b_2 \cdot 1_F)^{-1}(b_1 \cdot 1_F)^{-1}\left((a_1 \cdot 1_F)(b_2 \cdot 1_F) + (a_2 \cdot 1_F)(b_1 \cdot 1_F)\right)$$

$$= (b_1 \cdot 1_F)^{-1}(a_1 \cdot 1_F) + (b_2 \cdot 1_F)^{-1}(a_2 \cdot 1_F)$$

$$= \alpha\left(\frac{a_1}{b_1}\right) + \alpha\left(\frac{a_2}{b_2}\right)$$

$$\alpha\left(\frac{a_1}{b_1} \cdot \frac{a_2}{b_2}\right) = \alpha\left(\frac{a_1 a_2}{b_1 b_2}\right) = (b_1 b_2 \cdot 1_F)^{-1}(a_1 a_2 \cdot 1_F)$$

$$= \left((b_1 \cdot 1_F)(b_2 \cdot 1_F)\right)^{-1}\left((a_1 \cdot 1_F)(a_2 \cdot 1_F)\right)$$

$$= (b_2 \cdot 1_F)^{-1}(b_1 \cdot 1_F)^{-1}(a_1 \cdot 1_F)(a_2 \cdot 1_F)$$

$$= (b_1 \cdot 1_F)^{-1}(a_1 \cdot 1_F)\,(b_2 \cdot 1_F)^{-1}(a_2 \cdot 1_F)$$

$$= \alpha\left(\frac{a_1}{b_1}\right)\alpha\left(\frac{a_2}{b_2}\right)$$

$\alpha(1_{\mathbb{Q}}) = \alpha\left(\frac{1}{1}\right) = (1\cdot 1_F)^{-1}(1\cdot 1_F) = 1_F^{-1}\cdot 1_F = 1_F\cdot 1_F = 1_F$

$\therefore \alpha: \mathbb{Q} \longrightarrow F$是一个环同态.

$\because \mathbb{Q}$是域，F是非零环，$\alpha: \mathbb{Q} \to F$是一个环同态

$\therefore \alpha: \mathbb{Q} \to F$是一个单同态. $\therefore \mathbb{Q} \hookrightarrow F$

$\because$ 对$\forall x \in \mathbb{Z}$，$(\alpha\circ f)(x) = \alpha(f(x)) = \alpha\left(\frac{x}{1}\right)$

$= (1\cdot 1_F)^{-1}(x\cdot 1_F) = 1_F^{-1}(x\cdot 1_F) = 1_F(x\cdot 1_F) = x\cdot 1_F = \eta_F(x)$

$\therefore$ 下图交换：

$$\begin{tikzcd} & \mathbb{Z} \arrow[dl, "f"'] \arrow[dr, "\eta_F"] & \\ \mathbb{Q} \arrow[rr, "\alpha"'] & & F \end{tikzcd}$$

$\therefore \alpha = \theta$.

$\because \mathbb{Q}$是域，F是域，$\alpha: \mathbb{Q} \to F$是一个环同态

$\therefore \alpha(\mathbb{Q})$是$F$的子域

设K是F的一个任意的子域，$\because K$是F的子环 $\therefore 1_F \in K$.

对$\forall n \in \mathbb{Z}_{\geq 1}$，有：$n\cdot 1_F = \underbrace{1_F + \cdots + 1_F}_{n个1_F} \in K$.

对于$n=0$，有：$n\cdot 1_F = 0\cdot 1_F = 0_F \in K$

对$\forall n \in \mathbb{Z}_{\leq -1}$，有：$n\cdot 1_F = (-(-n))\cdot 1_F = -((-n)\cdot 1_F)$

$= -(\underbrace{1_F + \cdots + 1_F}_{(-n)个1_F}) \in K$

$\therefore$ 对$\forall n \in \mathbb{Z}$，有：$n\cdot 1_F \in K$.

对 $\forall \frac{a}{b} \in \mathbb{Q}$ (其中 $a \in \mathbb{Z}$, $b \in \mathbb{Z}\setminus\{0\}$).

假设 $b \cdot 1_F = 0_F$, 则 $\because F$ 是域, $\mathrm{char}(F) = 0 \in \mathbb{Z}_{\geq 0}$, $b \in \mathbb{Z}\setminus\{0\}$,

$b \cdot 1_F = 0_F$ $\quad \therefore \mathrm{char}(F) \mid b$ $\quad \therefore 0 \mid b$, $b \neq 0$ 矛盾.

$\therefore b \cdot 1_F \neq 0_F$ $\quad \therefore b \cdot 1_F \in F\setminus\{0_F\} = F^\times$ $\quad \therefore (b \cdot 1_F)^{-1} \in F^\times$ 且 $(b \cdot 1_F)^{-1} \in F$

$\because a \in \mathbb{Z}$ $\quad \therefore a \cdot 1_F \in K$

$\because b \in \mathbb{Z}\setminus\{0\}$ $\therefore b \cdot 1_F \in K$ $\quad \therefore b \cdot 1_F \in K\setminus\{0_F\} = K^\times$

$\therefore (b \cdot 1_F)^{-1} \in K$

$\therefore \alpha\left(\frac{a}{b}\right) = (b \cdot 1_F)^{-1}(a \cdot 1_F) \in K$ $\quad \therefore \alpha(\mathbb{Q}) \subseteq K$

$\therefore \alpha(\mathbb{Q})$ 是 F 的最小的子域.

$\because \alpha : \mathbb{Q} \longrightarrow \alpha(\mathbb{Q})$ 是环同构 $\quad \therefore \mathbb{Q} \simeq \alpha(\mathbb{Q})$

$\therefore \alpha(\mathbb{Q})$ 是 F 的最小的子域 且 $\mathbb{Q} \simeq \alpha(\mathbb{Q})$.

我已经证明了如下的引理:

Lemma: $\mathbb{Z}$ 是整环, $\mathbb{Q}$ 是域, $f : \mathbb{Z} \longrightarrow \mathbb{Q}$, $x \longmapsto \frac{x}{1}$ 是单同态,

F 是域且 $\mathrm{char}(F) = 0$, 则 $\iota_F : \mathbb{Z} \to F$, $x \mapsto x \cdot 1_F$ 是单同态, 且存在唯一的

环同态 $\theta : \mathbb{Q} \longrightarrow F$, $\frac{a}{b} \longmapsto (b \cdot 1_F)^{-1}(a \cdot 1_F)$, s.t. 下图交换:

$$\begin{tikzcd} & \mathbb{Z} \arrow[dl, "f"'] \arrow[dr, "\iota_F"] & \\ \mathbb{Q} \arrow[rr, "\theta"'] & & F \end{tikzcd}$$

$\theta : \mathbb{Q} \to F$ 是一个单同态, 且 $\theta(\mathbb{Q})$ 是 F 的

最小的子域, 且 $\mathbb{Q} \simeq \theta(\mathbb{Q})$.

Lemma：F是一个任意的域，存在环同态 $\alpha:\mathbb{Q}\to F$，则有：$\mathrm{char}(F)=0$ 且 $\alpha:\mathbb{Q}\to F$ 是单同态.

Proof：对 $\forall n\in\mathbb{Z}_{\geq 1}$，有：

$$\alpha\left(\frac{n}{1}\right)=\alpha\Big(\underbrace{\frac{1}{1}+\cdots+\frac{1}{1}}_{n个\frac{1}{1}}\Big)=\underbrace{\alpha\left(\frac{1}{1}\right)+\cdots+\alpha\left(\frac{1}{1}\right)}_{n个\alpha(\frac{1}{1})}$$

$$=\underbrace{1_F+\cdots+1_F}_{n个1_F}=n\cdot 1_F$$

$$\alpha\left(\frac{0}{1}\right)=\alpha(0_{\mathbb{Q}})=0_F=0\cdot 1_F$$

对 $\forall n\in\mathbb{Z}_{\leq -1}$，有：

$$\alpha\left(\frac{n}{1}\right)=\alpha\left(\frac{-(-n)}{1}\right)=\alpha\left(-\frac{-n}{1}\right)=-\alpha\left(\frac{-n}{1}\right)=-\left((-n)\cdot 1_F\right)$$

$$=n\cdot 1_F$$

$\therefore$ 对 $\forall n\in\mathbb{Z}$，有：$\alpha\left(\frac{n}{1}\right)=n\cdot 1_F$

假设 $\mathrm{char}(F)\neq 0$，则有：$\mathrm{char}(F)$是素数. 设 $\mathrm{char}(F)=p$，$p\in\mathbb{Z}_{\geq 2}$ 且 p 是素数.

$\because F$是域 $\therefore F$是整环. $\because \mathrm{char}(F)=p\in\mathbb{Z}_{\geq 2}$，对于 $p\in\mathbb{Z}_{\geq 2}$，有：$\mathrm{char}(F)\mid p$ $\therefore p\cdot 1_F=0_F$. $\therefore \alpha\left(\frac{p}{1}\right)=p\cdot 1_F=0_F$.

$\because \frac{p}{1}\in\mathbb{Q}\setminus\{0_{\mathbb{Q}}\}=\mathbb{Q}^{\times}$ $\therefore \alpha\left(\frac{p}{1}\right)\in F^{\times}=F\setminus\{0_F\}$.

$\therefore 0_F\in F\setminus\{0_F\}$ $\therefore 0_F\notin\{0_F\}$. 矛盾. $\therefore \mathrm{char}(F)=0$

$\because$ $\mathbb{Q}$是域，F是域，F是非零环，$\alpha: \mathbb{Q} \to F$是环同态

$\therefore$ $\alpha: \mathbb{Q} \to F$是单同态. $\square$